

# This Month's DVD Contents

This month's DVD is a compilation of different tools starting from integrated development for Intel Galileo to circuit compilers like MikroElektronika, along with simulators like quite universal circuit simulator (Qucs). Also, enjoy the rhythm platform through Performous

## SNEHA AMBASTHA

### PICBASIC PRO

PICBASIC PRO is a simple and easy-to-use Basic compiler with dedicated libraries or routines for the rapid development of Microchip PIC microcontroller (MCU) based projects. It produces codes in the same manner as a C compiler and so, is widely used by engineering professionals to produce commercial firmware. Its design for fast compilation makes it popular among professionals.

### Arduino IDE for Intel Galileo

The integrated development environment (IDE) for Intel Galileo board, with x86 architecture, is compatible with operating systems like Windows (7, 8, 8.1), Ubuntu (12.04) and Mac (OS X version 10.8.5, OS X 10.6.8, 10.7.5 and 10.9.2). It supports Arduino libraries for the following: serial peripheral interface (SPI), electrically-erasable programmable read-only memory (EEPROM), universal asynchronous receiver/transmitter (UART), general-purpose input/output (GPIO), Wi-Fi, Servo and USB host to help board interfaces.

### mikroC PRO

mikroC PRO is an ANSI C compiler for PIC microcontroller based devices from Microchip. It has an IDE, a compiler with advanced optimisations, both hardware and software libraries and additional help tools. It also has some ready-to-use examples for users to get started in no time.

### mikroBasic PRO

mikroBasic PRO is a full-featured Basic compiler for PIC development from Microchip, built specially for beginners with its simple syntax and clear code.

It has over 500 ready-to-use library functions and examples, along with a comfortable IDE and comprehensive help file to provide sufficient aid in development.

### mikroPascal PRO

mikroPascal PRO is a full-featured Pascal compiler with some ready-to-use libraries, intuitive IDE and compiler with advanced optimisations for PIC devices from Microchip. It features about four different levels of optimisations that can reduce the code size up to 20 per cent.

### Quite universal circuit simulator (Qucs)

Qucs is an integrated circuit simulator with a graphical user interface (GUI) to simulate a large-signal, small-signal as well as noise behaviour, of the circuit. It aims to support all kinds of circuit simulation types like AC, DEC, harmonic balance analysis, noise analysis and S-parameter. It can also import existing SPICE models for use in simulations.

### SmartSim

SmartSim is an open source, cross-platform digital logic circuit design and simulation package for educational use for anyone, from children to adults. It helps create complex circuits by allowing one to create one's own custom components and include them in other circuits. It also provides many semiconductor based components and models such as opamps, diodes, MOSFETs and PMOSFETs.

### QP Modeler

Quantum platform (QP) modeler (version 3.2.2), known as QM, is a graphical modelling tool used to design and implement real-time embedded software based

## Popular resources

**7Zip.** Version 9.20 (file archiver)  
**Java Runtime Environment.** Version 8.0.250.18 (runs programs written in Java programming language)  
**VLC.** Version 2.1.5 (media player)  
**Opera.** Version 24.0.1558.66 (web browser)  
**Free Download Manager.** Version 3.9.4 (file download manager)  
**Thunderbird.** Version 31.1.0 (e-mail application)  
**Mozilla Firefox.** Version 33.1 (web browser)  
**Apache Open Office.** Version 4.1.1 (open office)  
**Prozek Vital 2014** (antivirus)

on unified modelling language (UML) state machines and lightweight QP active object frameworks. It helps one to decompose software into event-driven components called active objects and to automatically generate C or C++ codes.

### ZenitPCB suite 180

ZenitPCB suite is a printed circuit board (PCB) layout tool to help hobbyists, students and academics from high schools create their own PCBs with a professional approach, without an expensive licence. It is a computer aided design (CAD) program that helps one to realise his or her projects in a short time.

### Wiring

Wiring is a programming framework for microcontrollers to allow writing cross-platform software to control devices attached to a wide range of microcontroller boards and to create all kinds of creative coding, interactive objects, spaces or physical experiences. Currently, it is being used widely by thousands of students, artists, designers, researchers and hobbyists for learning, prototyping and finished professional work production.

Some others tools are: ScadaBR, Modbus SCADA, timer calculator V2.7.0, package manager V3.5.0, visual TFT V3.8.0, visual GLDC V2.7.0, GLCD font creator V1.2.0 and Smith V3.10. ●

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